



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

NeuroText: Intelligent Semantic Analysis for Detecting Plagiarism and Paraphrased Content



TEAM MEMBERS

Madhu Sneha(192425092)
G. DeviSri(192472025)



PROJECT GUIDE

Dr. T. Kumaragurubaran



SUBJECT DETAILS

DSA0306
Natural Language Processing

Problem Background



Application Domain

Natural Language Processing (NLP) — analyzing text content using AI to understand sentence meaning and detect semantic similarities between documents.



Current Scenario

Traditional plagiarism checkers mainly focus on **matching words or phrases** and may fail to detect content that has been **reworded or paraphrased**.



Need for the Project

There is a need for an intelligent system that can understand the **meaning of sentences**, identify semantically similar content, and detect plagiarism even when the wording is different.



Target Users

Students, teachers, researchers, content creators, and educational institutions who need to verify the originality of written content.

Problem Statement

- Traditional plagiarism detection systems mainly depend on exact word or phrase matching. They often fail to identify paraphrased or semantically similar content, where the same idea is expressed using different words.

LIMITATIONS OF EXISTING SYSTEMS

Depend mainly on exact word matching

⚠ Fail to detect paraphrased sentences effectively

⚠ Cannot understand the deeper semantic meaning of text

⚠ Produce inaccurate results when sentence structure or vocabulary is changed

Objectives



PRIMARY OBJECTIVE

Develop an intelligent NLP-based system that detects plagiarism and paraphrased content by understanding the semantic meaning of sentences.

01



Text Processing

Clean, normalize, and divide the original and suspected documents into individual sentences.

02



Embedding Generation

Generate Sentence Embeddings
Use SBERT (Sentence-BERT) to convert sentences into meaningful numerical representations.

03



Similarity Detection

Use Cosine Similarity to compare sentence embeddings and identify semantically similar content.

04



Content Classification

Classify the content as **Original, Plagiarized, or Paraphrased** based on similarity scores.

Project Scope: A working NLP-based application covering **text input, preprocessing, SBERT embedding generation, cosine similarity analysis, classification, and detection report generation** for plagiarism and paraphrased content.

Existing & Proposed System

EXISTING SYSTEM



Mainly depends on exact word or keyword matching



Struggles to detect paraphrased content



Limited understanding of sentence meaning



May produce inaccurate plagiarism results

PROPOSED SYSTEM



Uses **SBERT** to understand sentence meaning



Generates **sentence embeddings** for semantic comparison

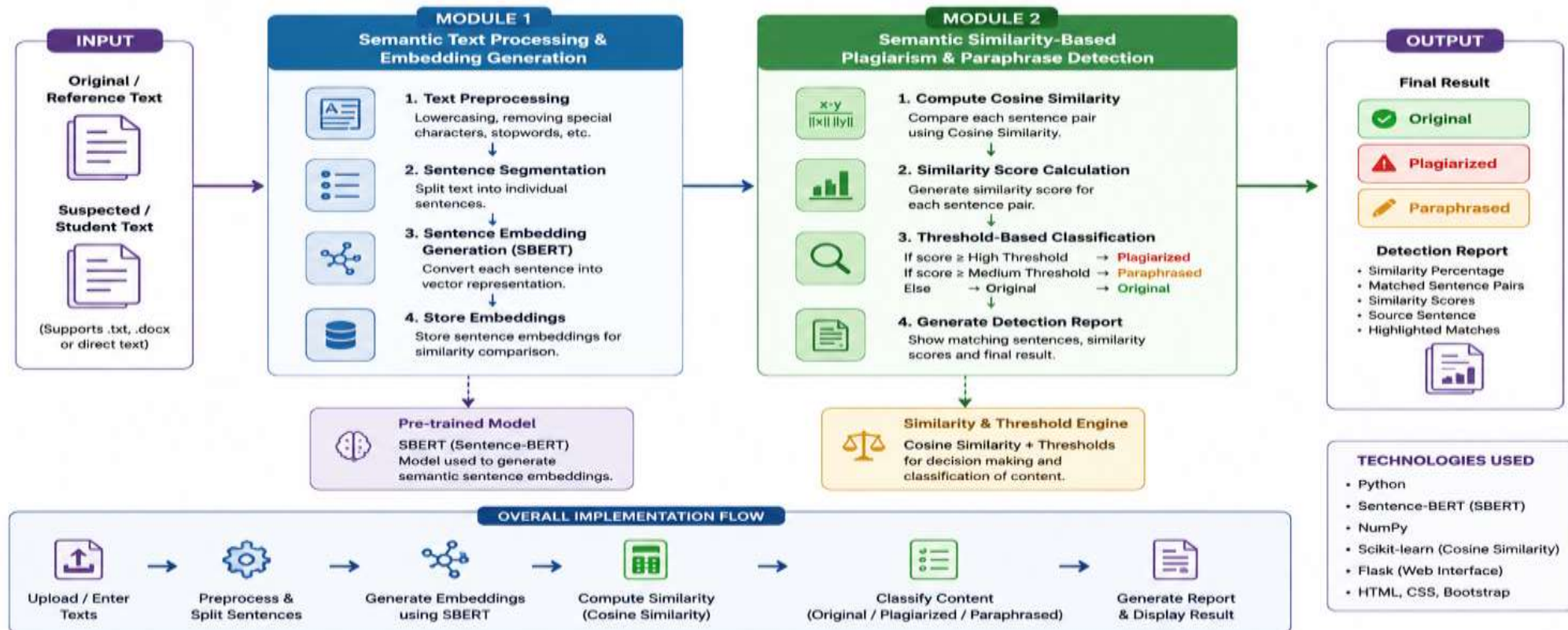


Uses **Cosine Similarity** to detect similar meaning



Classifies content as **Original, Plagiarized, or Paraphrased**

ARCHITECTURE DIAGRAM



LITERATURE SURVEY

Author and Year	Focus	Key Findings	Research Gaps
Vrbanec and Meštrović (2023)	Unsupervised paraphrase detection using deep learning and semantic similarity techniques.	Deep learning models showed better performance than traditional statistical similarity methods for detecting paraphrased content.	Further improvement is needed for large-scale document-level plagiarism detection and real-time analysis.
Authors of Optimized Algorithm Design (2023)	Text similarity detection using AI, NLP, TF-IDF, Cosine Similarity, SimCSE, and SBERT.	Combining multiple similarity models improved accuracy while reducing computational requirements.	The approach requires further validation for complex plagiarism and highly paraphrased content.
Xiong et al. (2024)	BERT-based plagiarism detection using reinforcement learning and optimized pretraining techniques.	The proposed approach improved plagiarism detection performance and handled minority plagiarism classes effectively.	High computational complexity may limit deployment in lightweight and real-time systems.
Abisheka, Deisy and Sharmila (2024)	Transformer-based semantic relation extraction for contextual paraphrased plagiarism detection.	The system improved detection of semantic manipulation and structural reorganization using transformer-based analysis.	Scalability and broader multilingual evaluation require further research.
Sajid et al. (2025)	Comparative analysis of text-based plagiarism detection techniques.	Advanced semantic analysis and hybrid approaches were identified as promising methods for detecting highly disguised plagiarism.	More research is needed to develop robust systems that combine multiple techniques efficiently.

ALGORITHM USED

ALGORITHM USED

- Sentence-BERT (SBERT) is a Transformer-based NLP algorithm used to generate meaningful sentence embeddings.
- It converts sentences into numerical vector representations while preserving their semantic meaning.
- It is used in our system for semantic text analysis and similarity detection.
- It can identify similarities between sentences even when they use different words or sentence structures.
- Provides effective semantic understanding, accuracy, and efficient text comparison.
- The generated embeddings help detect plagiarized and paraphrased content based on semantic similarity.

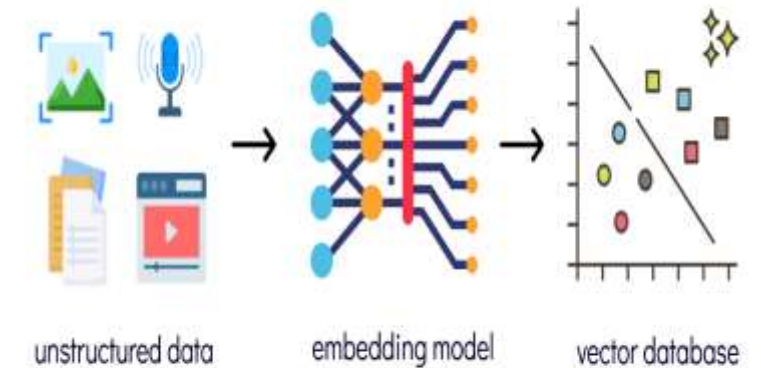
TECHNIQUES USED

- Sentence-BERT (SBERT): Generates semantic embeddings and understands the contextual meaning of sentences.
- Cosine Similarity: Measures the similarity between sentence embedding vectors.
- Text Preprocessing: Cleans, normalizes, and divides the input text into individual sentences.
- Similarity-Based Classification: Classifies the content as Original, Plagiarized, or Paraphrased based on similarity scores.

MODULE 1

Semantic Text Processing & Embedding Generation

- Semantic text processing deals with understanding the **meaning of text**.
- It helps computers understand the **relationship between words and sentences**.
- Context is important for determining the meaning of a word.
- **Tokenization** divides text into smaller units called tokens.
- **Word Sense Disambiguation** identifies the correct meaning of an ambiguous word.
- **Semantic similarity** measures the similarity between two pieces of text.
- **Text embeddings** represent text as numerical vectors.



Sentence:

“Priya sat on the bank and watched the boats.”

OUTPUT

NEUROTTEXT

INTELLIGENT SEMANTIC ANALYSIS

01 TEXT INPUT & PREPROCESSING

02 SBERT EMBEDDINGS

03 SEMANTIC MATCHING

04 CLASSIFICATION

05 RESULTS & REPORT

SHOW ALL PHASES (FULL DASHBOARD)

ANALYSIS ENGINE READY

NeuroText

Intelligent Semantic Analysis for Detecting Plagiarism and Paraphrased Content

SYSTEM STATUS: READY

01 — TEXT INPUT & PREPROCESSING

Provide the original document and suspected document for semantic analysis.

ORIGINAL DOCUMENT

Paste Text

Upload .txt file

Paste the original document here...

CHARACTER COUNT

WORD COUNT

SENTENCES

0

0

0

SUSPECTED DOCUMENT

Paste Text

Upload .txt file

Paste the suspected document here...

CHARACTER COUNT

WORD COUNT

SENTENCES

0

0

0

Advanced Settings: Classification Thresholds

RUN COMPLETE NEUROTTEXT PIPELINE (PHASES 1-5)

NEUROTTEXT

INTELLIGENT SEMANTIC ANALYSIS

01 TEXT INPUT & PREPROCESSING

02 SBERT EMBEDDINGS

03 SEMANTIC MATCHING

04 CLASSIFICATION

05 RESULTS & REPORT

SHOW ALL PHASES (FULL DASHBOARD)

ANALYSIS ENGINE READY

NeuroText

Intelligent Semantic Analysis for Detecting Plagiarism and Paraphrased Content

SYSTEM STATUS: READY

02 — SBERT EMBEDDING GENERATION

Batch encoding sentences into 384-dimensional dense vectors using Sentence-BERT.

MODEL

all-MiniLM-L6-v2

EMBEDDING DIMENSION

384

ORIGINAL SHAPES

(5, 384)

SUSPECTED SHAPES

(5, 384)

Inspect Embedding Vector Sample (First 5 Dimensions)

	Dim 0	Dim 1	Dim 2	Dim 3	Dim 4	Dim 5
Suspected Sentence #0		0.0484	0.0919	0.3986	-0.001	0.0961
Suspected Sentence #1		0.0050	0.0056	-0.0435	-0.0189	0.0532
Suspected Sentence #2		-0.0636	-0.0033	-0.0752	-0.0666	0.0539
Suspected Sentence #3		0.0447	0.0612	0.1077	0.005	-0.028
Suspected Sentence #4		0.0941	0.0314	0.0233	-0.006	0.0057

MODULE 2

Semantic Similarity & Plagiarism Detection

- Semantic similarity measures the similarity between two texts based on their meaning.
- It helps identify whether two sentences express the same idea.
- Text is converted into numerical vectors called embeddings.
- The embeddings of two texts are compared to find their similarity.
- Cosine similarity is commonly used to measure similarity.
- Plagiarism detection identifies copied or similar content.



Engineer to Excel



OUTPUT

NEUROTTEXT

INTELLIGENT SEMANTIC ANALYSIS

01 TEXT INPUT & PREPROCESSING

02 SBERT EMBEDDINGS

03 SEMANTIC MATCHING

04 CLASSIFICATION

05 RESULTS & REPORT

SHOW ALL PHASES (FULL DASHBOARD)

ANALYSIS ENGINE READY

03 — SEMANTIC SIMILARITY & MATCHING

Pairwise Cosine Similarity matrix calculation and best-match identification.

PAIRWISE COSINE SIMILARITY MATRIX

	Original 1	Original 2	Original 3	Original 4	Original 5
Suspected 1	1	0.3322	0.4423	0.0378	0.0528
Suspected 2	0.3771	1	0.8743	0.4275	0.0354
Suspected 3	0.4045	0.4025	1	0.9797	0.0026
Suspected 4	0.0378	-0.0227	0.1196	1	0.1404
Suspected 5	0.0528	-0.0311	-0.0184	0.0012	1

BEST SENTENCE MATCHING TABLE

Suspected #	Suspected Sentence	Best Original Match
1	Artificial intelligence is transforming modern education.	Artificial intelligence is transforming modern education.
2	Digital learning platforms give learners flexible access to educational resources.	Online learning platforms provide students with flexible access to educational resources.
3	Machine learning can assist teachers in identifying students who require additional support.	Machine learning can help teachers identify students who need additional support.
4	Regular exercise improves physical health and mental well-being.	Regular exercise improves physical health and mental well-being.
5	The capital city of France is Paris.	Artificial intelligence is transforming modern education.

NEUROTTEXT

INTELLIGENT SEMANTIC ANALYSIS

01 TEXT INPUT & PREPROCESSING

02 SBERT EMBEDDINGS

03 SEMANTIC MATCHING

04 CLASSIFICATION

05 RESULTS & REPORT

SHOW ALL PHASES (FULL DASHBOARD)

ANALYSIS ENGINE READY

04 — CONTENT CLASSIFICATION & SUSPICIOUS TEXT DETECTION

Evaluating similarity scores against configurable project thresholds to isolate suspicious content.

TOTAL SENTENCES
5

ORIGINAL
1 (20.00%)

PARAPHRASED
0 (0.00%)

PLAGIARIZED
4 (80.00%)

SUSPICIOUS CONTENT CARDS

Sentence #1
Suspected Content: Artificial intelligence is transforming modern education.
Best Original Match: Artificial intelligence is transforming modern education.
Similarity: 100.00%
PLAGIARIZED

Sentence #2
Suspected Content: Digital learning platforms give learners flexible access to educational resources.
Best Original Match: Online learning platforms provide students with flexible access to educational resources.
Similarity: 87.43%
PLAGIARIZED

Sentence #3
Suspected Content: Machine learning can assist teachers in identifying students who require additional support.
Best Original Match: Machine learning can help teachers identify students who need additional support.
Similarity: 97.97%
PLAGIARIZED

Sentence #4
Suspected Content: Regular exercise improves physical health and mental well-being.
Best Original Match: Regular exercise improves physical health and mental well-being.
Similarity: 100.00%
PLAGIARIZED

RESULTS

NEUROTTEXT

INTELLIGENT SEMANTIC ANALYSIS

01 TEXT INPUT & PREPROCESSING

02 SBERT EMBEDDINGS

03 SEMANTIC MATCHING

04 CLASSIFICATION

05 RESULTS & REPORT

SHOW ALL PHASES (FULL DASHBOARD)

ANALYSIS ENGINE READY

05 — RESULTS DASHBOARD & DETECTION REPORT

Executive summary, distribution chart, highlighted document analysis, and downloadable PDF/CSV reports.

Overall Semantic Similarity

78.45%

The score represents the average semantic similarity between suspected sentences and their best matching original sentences.

DOCUMENT STATUS

HIGH SIMILARITY

High semantic similarity detected.
Potentially suspicious content requires
academic review.

CONTENT CLASSIFICATION DISTRIBUTION



CONCLUSION

- Neuro Text provides an intelligent solution for detecting both plagiarism and paraphrased content using NLP techniques.
- The system uses Sentence-BERT (SBERT) to generate semantic embeddings and understand the contextual meaning of text.
- Cosine Similarity is used to compare text embeddings and identify the level of semantic similarity between documents.
- The system can classify content as Original, Plagiarized, or Paraphrased, improving traditional plagiarism detection methods.
- In the future, Neuro Text can be enhanced with multilingual support, larger datasets, advanced AI models, and real-time document analysis.

REFERENCES

- [1] T. Vrbanec and A. Meštrović, “Comparison study of unsupervised paraphrase detection: Deep learning—The key for semantic similarity detection,” *Expert Systems*, vol. 40, no. 9, 2023,
- [2] J. Latina, E. D. M. Vallejo, G. M. Cabalsi, and C. J. Centeno, “Utilization of NLP Techniques in Plagiarism Detection System through Semantic Analysis using Word2Vec and BERT,” in *2024 International Conference on Expert Clouds and Applications (ICOECA)*, 2024,
- [3] J. Xiong *et al.*, “Efficient reinforcement learning-based method for plagiarism detection boosted by a population-based algorithm for pretraining weights,” *Expert Systems with Applications*, vol. 238, 2024,
- [4] D. Koka, I. V. Krishna, M. V. Rao, S. V. Inukurthi, and M. A. Babu, “Plagiarism Detection Using NLP and Semantic Similarity Techniques,” 2025.
- [5] “NLP-Driven Plagiarism Detection: A Smarter Approach,” in *2025 2nd International Conference on Integration of Computational Intelligent System (ICICIS)*, 2025,

Thank You